Lab experiment

2.Identify the system calls to copy the content of one file to another and illustrate the same using a C program.

Aim:

To identify and use the appropriate system calls to copy the contents of one file into another file using a C program.

Algorithm:

Include necessary headers:
Include the necessary header files like <stdio.h>, <fcntl.h>, and <unistd.h> for working with file-related system calls.
Declare variables:
Declare variablesto hold file descriptors, buffer, and other necessary information.
Open the source and destination files:
Use the open system call to open the source and destination files,obtaining file descriptors for each.
Read from the source file:
Use the read system call to read data from the source file into abuffer.
Write to the destination file:
Use the write system call to write the data read from the source fileinto the destination file.
Close the files:
Use the close system call to close the source and destination files.

Coding:

```
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
   int source, destination;
   char buffer[100];
   ssize_t bytes;

   source = open("source.txt", O_RDONLY);

   if (source < 0)
```

```
{
    perror("Error opening source file");
    return 1;
}

destination = open("destination.txt",
                   O_WRONLY | O_CREAT | O_TRUNC,
                   0644);

if (destination < 0)
{
    perror("Error opening destination file");
    close(source);
    return 1;
}

printf("File copy program\n");

while ((bytes = read(source, buffer, sizeof(buffer))) > 0)
{
    ssize_t total = 0;

    while (total < bytes)
    {
        ssize_t written = write(destination,
                                buffer + total,
                                bytes - total);

        if (written < 0)
        {
            perror("Error writing file");
            close(source);
            close(destination);
            return 1;
        }

        total += written;
    }
}

if (bytes < 0)
{
    perror("Error reading file");
    close(source);
```

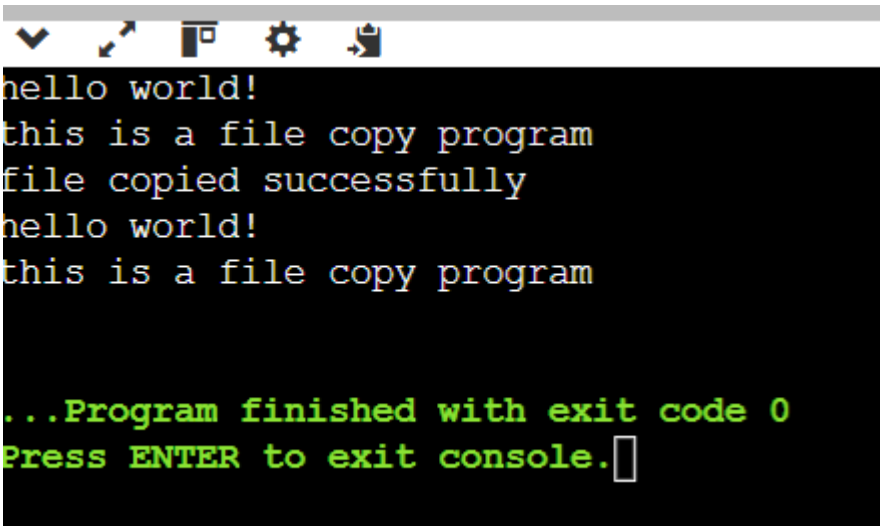
```
        close(destination);
        return 1;
    }

    close(source);
    close(destination);

    printf("File copied successfully\n");

    return 0;
}
```

Output:



Result:

Thus, the contents of one file were successfully copied to another file using the `open()`, `read()`, `write()`, and `close()` system calls in C.